



Straight from the Horse's Mouth

Where the Wild Things Aren't

Student Handout

Name: _____

Objective

Upon completing this lesson, you will be able to:

- interpret a text in order to justify the role of genetic variation, genetic drift, and bottlenecks in evolution and extinction.

Directions

Read the article "Scientists Clone an Endangered Przewalski's Horse for the First Time, And It's So Cute". Once you have read it, use the text in order to answer the questions below.

1) What were the major causes of the decline of the population of native Przewalski's horses? How many survived in captivity? Why is this a problem for the long-term success of the Przewalski's horse?



Figure 1: Przewalski's horse is genetically different from modern domestic horses, and neither is an ancestor of the other.

2) What is a bottleneck? How does a bottleneck event affect a population's gene pool?

3) What is genetic drift? How does genetic drift affect genetic diversity?

4) Why is low genetic variability concerning to conservation biologists?

5) One way conservation biologists at the San Diego Zoo is by cloning Kuporovic, a Przewalski's horse that died more than 20 years ago. Why did scientists clone this specific horse?

b) How will this colt, Kurt, help to preserve the species? Relate your answer to natural selection and evolution.